

# Stepper Motor Drive Guide for AMR Telescopic Slides

This guide explains how to operate, configure, and manage a telescopic slide mechanism using a stepper motor. It details low-level hardware control, comparison of drive types, microcontroller logic, and compares it to how Tusk Robots manages its telescoping tynes.

## 1. How to Operate a Stepper Motor (Low-Level Control)

To operate a stepper motor on an AMR, you need three core components in the electronic control stack:



### A. Closed-Loop vs. Open-Loop Stepper Systems

- **Open-Loop (Not Recommended for AMRs):** Simple and cheap, but if the forks encounter an obstruction (e.g., hitting a pallet wall), the motor will lose steps, causing the navigation system to lose track of the fork extension distance.
- **Closed-Loop (Recommended):** Utilizes a NEMA 23/24 stepper motor with an **integrated rotary encoder** (typically 1000 to 4000 CPR). The driver monitors the encoder feedback and automatically adjusts current to prevent step loss and can signal a fault to the main controller if a stall occurs.

### B. Controller Interface Methods

#### 1. Pulse & Direction (PUL / DIR):

- The microcontroller sends a square-wave frequency to the driver's **PUL** pin (1 pulse = 1 microstep rotation) and toggles the **DIR** pin high/low to control direction.

- A third **ENA** (Enable) pin is toggled to power on/off the motor coils.

## **2. Fieldbus Control (CANopen / Modbus RTU):**

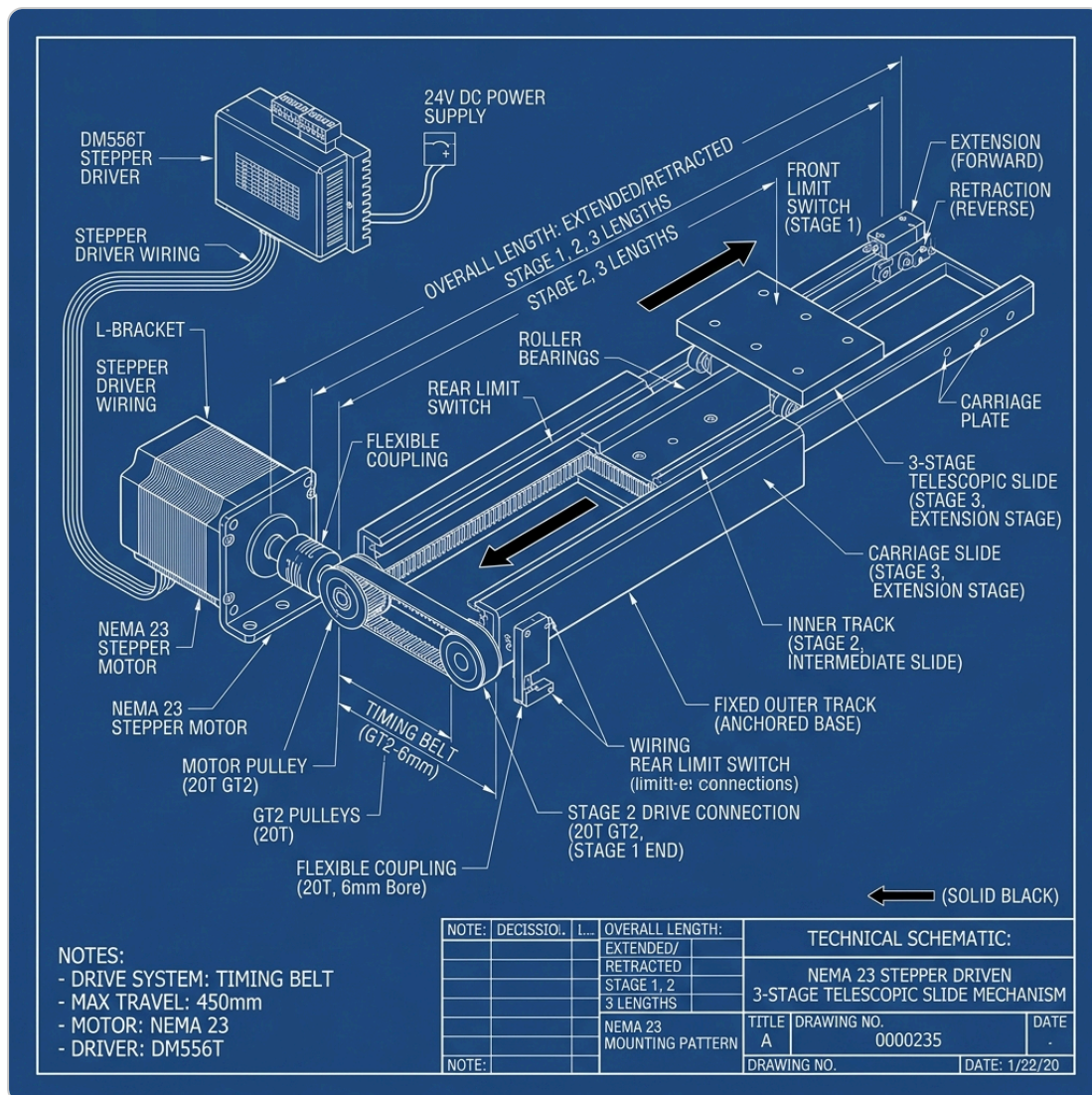
- Uses serial communication lines to send command packets (e.g., "Go to position 5000 steps at speed 1000 RPM"). This eliminates pulse wiring and allows the controller to read real-time torque and speed values from the driver.

## **2. Stepper Drive Configurations (Types of Drives)**

### **Option A: Timing Belt & Pulley Drive**

The stepper motor is mounted at the rear of the slide and rotates a driving pulley, which pulls a loop of heavy-duty timing belt (e.g., GT2 6mm or 9mm width) attached to the moving stages.

- **Diagram:**



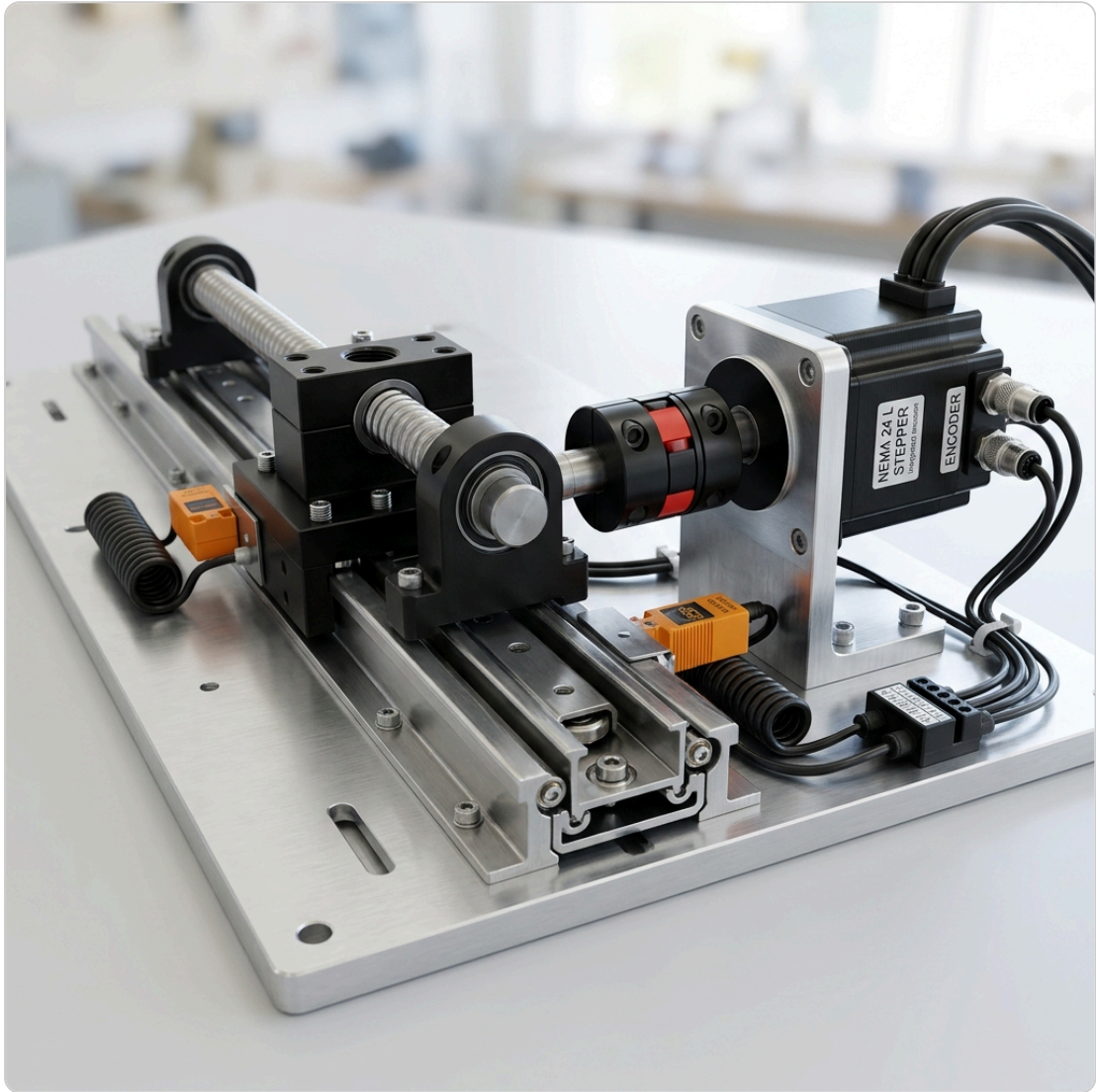
*NEMA 23 Stepper Driven 3-Stage Telescopic Slide Timing Belt System (Illustrative Schematic / Non-Realistic)*

- **Key Specifications:**
- **Stepper Motor:** NEMA 23 High-Torque (  $1.8^\circ$  step angle, 2.2 Nm holding torque)
- **Stepper Driver:** DM556T (24V-50V DC power supply, set to 1600 steps/revolution microstepping)
- **Drive Pulley:** 20-Tooth GT2 Pulley (  $\varnothing 6.35 \text{ mm}$  bore to match motor shaft)
- **Performance:** High extension speed, quiet operation, but susceptible to belt stretch under heavy load.

## Option B: Ball Screw Drive

The stepper motor is coupled directly to a precision ball screw (e.g., SFU1605) using a flexible jaw coupling. As the screw rotates, the ball nut drives the intermediate stage.

- CAD Render:



*NEMA 24 Closed-Loop Stepper coupled to Ball Screw Telescopic Assembly (Realistic 3D CAD Render)*

- **Key Specifications:**
- **Stepper Motor:** NEMA 24 Closed-Loop Stepper with Integrated Encoder
- **Drive System:** SFU1605 Ball Screw (  $16\text{ mm}$  diameter,  $5\text{ mm}$  lead)
- **Coupling:** Flexible D30L40 Plum Coupling (  $8\text{ mm}$  to  $10\text{ mm}$  bore)

- **Limit Switches:** Inductive Proximity Sensors (LJ12A3-4-Z/BX) at both travel limits
- **Performance:** Extreme thrust force, zero back-driving, high precision, but slower extension speed.

### 3. Low-Level Control Logic (STM32/Arduino Pseudo-code)

Below is the control logic for homing, extending, and detecting obstructions using a pulse-generator microcontroller:



```
// Check for safety limit switch or driver fault
if (digitalRead(limitExtended) == LOW || digitalRead(driverFault) == LOW)
    stopMotor();
    break;
}

// Pulse output (implementing simple ramp profile for acceleration)
int pulseDelay = calculateRampDelay(i, targetSteps);
digitalWrite(pulPin, HIGH);
delayMicroseconds(pulseDelay);
digitalWrite(pulPin, LOW);
delayMicroseconds(pulseDelay);
}
}
```

## 4. How Telescoping Slides are Managed in Tusk Robots

In official commercial Tusk AMRs (such as the E10T telescopic fork model), the telescoping mechanism is managed via a robust industrial architecture:

### 1. Integrated Geared Servos (Instead of Open-Loop Steppers):

Tusk uses high-efficiency **brushless DC geared servo motors** or **closed-loop hybrid step-servos** operating on a 24V or 48V DC bus. They are geared down (typically 1:10 or 1:15) to multiply torque for handling heavy pallets.

### 2. CANopen Network Interface:

The motor drives communicate with the low-level micro-ROS node (running on an STM32 MCU) via a **CANopen fieldbus** loop. This allows the robot to send precise velocity and position commands and read motor temperature, current, and encoder counts in real-time.

### 3. Dual Fork Synchronization:

The Tusk E10T uses a single motor linked to a transverse splined shaft that drives the left and right telescopic slides together. This mechanical lock ensures the forks extend in perfect synchronization, preventing binding.

#### **4. Current-Limit Obstruction Detection:**

The STM32 low-level safety controller constantly monitors the current consumption of the fork drive motor. If the current spikes above a safety threshold (indicating that the forks have collided with a pallet beam or obstruction), the MCU immediately commands a halt, cuts driver power, and triggers a ROS E-stop alert.

#### **5. Inductive Limit Proximity Sensors:**

PNP-type inductive proximity sensors are mounted at the retracted (home) and fully-extended limits inside the fork casing. They act as hard-stop electrical interlocks to prevent mechanical crash damage.